

HPIM-NoC: A Priori-Knowledge-Based Optimization Framework for Heterogeneous PIM-Based NoCs

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Abstract—Network-on-Chip (NoC) accelerators with heterogeneous Processing-in-Memory (PIM) cores achieve superior performance than homogeneous ones for neural networks. Dedicated simulators and architecture search frameworks are pivotal for obtaining performance, power, and area (PPA) metrics, as well as guiding the design process. However, existing simulators are primarily designed for homogeneous NoC and lack support for simulating heterogeneous PIM-based NoC architectures. Besides, current search frameworks for heterogeneous NoC architectures only focus on workload allocation and mapping strategies, failing to explore heterogeneous PIM configurations in a larger design space. In this work, we propose HPIM-NoC, a joint simulation and search framework for heterogeneous PIM-based NoC architectures. HPIM-NoC not only supports the simulation of heterogeneous PIM cores, but also provides more accurate latency results by introducing NoC transmission delays and pipelines in co-simulation. HPIM-NoC implements a three-stage heterogeneous search process based on priori knowledge and employs a specific simulated annealing algorithm tailored for heterogeneous architecture search. The search process is accelerated by precomputing core PPA metrics and reducing NoC simulation frequency. In addition, the framework integrates a customized layout algorithm to optimize the placement of heterogeneous NoC, minimizing communication latency and overall area. Experimental results on various neural networks demonstrate that HPIM-NoC can quickly find near-optimal configurations within a limited time. The proposed acceleration method reduces the search time of HPIM-NoC by 2.12 $\times$, 2.17 $\times$, and 2.96 $\times$, respectively. Compared to homogeneous architectures, the Fusions of Metrics (FoMs) of heterogeneous PIM-based NoC architectures found by HPIM-NoC are reduced by 1.18%, 16.94%, and 37.41% for ResNet-18 under three settings, respectively.

Index Terms—heterogeneous, PIM, NoC, simulation, architecture search

I. INTRODUCTION

In recent years, Convolutional Neural Networks (CNNs) have shown exceptional capabilities in applications such as object detection and face recognition. However, as CNN models grow more complex, traditional von-Neumann architectures, like CPUs and GPUs, face significant challenges due to the massive data movements between memory and computing units. The data movements consume over 80% of system energy and worsen the "memory wall" problem [1]. To address these challenges, Processing-In-Memory (PIM) architectures [2], [3] have been introduced, performing Matrix-Vector Multiplication (MVM) directly within memory arrays to eliminate data movement overhead and improve energy efficiency.

As the number of neural network parameters increases, single-core PIM accelerators face limitations in computation

capacity and still require weight refreshes with significant overhead [4]. Current approaches often design homogeneous or heterogeneous PIM-based NoC accelerators to store all weights on-chip. However, the performance of homogeneous PIM-based NoC architectures is tightly coupled to the PIM types and array sizes of individual cores, which restricts their scalability and flexibility [5]. SRAM-based digital PIM architectures with integrated logic computation [6], [7] exhibit low latency, while RRAM-based analog PIM architectures [8], [9] provide high computational density, despite the overhead of analog-to-digital conversion. Heterogeneous PIM-based NoC architectures, which combine different PIM types and core sizes, offer improved scalability and flexibility to meet stringent power, performance, and area (PPA) requirements. The main challenge in designing heterogeneous PIM-based NoC architectures lies in the vast design space, which makes manual design almost impossible and requires efficient design space exploration. The exploration time depends on the simulation per architecture and the search strategy. In terms of simulation, most existing multi-core simulators like DNN+NeuroSIM v2.0 [10] and MNSIM 2.0 [11] target homogeneous PIM-based architectures and do not support the simulation of heterogeneous architectures. Moreover, existing simulators [10], [11] primarily focus on simulating core-level metrics, lacking accurate models for simulating inter-core communication in NoC. Besides, general-purpose NoC simulators such as Booksim 2.0 [12] cannot model the inter-layer data dependencies of neural networks, resulting in inaccuracies in latency simulation outcomes.

In terms of architectural search, current architecture search frameworks face limitations due to the lack of simulators tailored for heterogeneous PIM-based NoC architectures, which hinders their ability to evaluate diverse hardware configurations efficiently. Frameworks such as PIM-HLS [13] and [14] typically focus on searching for workload allocation and mapping schemes rather than hardware configurations. Other frameworks, like Gibbon [15] and AIG-CIM [16], primarily search for homogeneous configurations or limited ranges of heterogeneous configurations, thus missing the opportunity to explore a wider design space involving multiple PIM types, core sizes, and array configurations. As a result, there is a critical need for dedicated search frameworks to find optimal configurations for heterogeneous PIM-based NoC architectures.

In this paper, we propose HPIM-NoC, a priori-knowledge-based optimization framework for heterogeneous PIM-based

NoC architectures. HPIM-NoC enables the simulation of heterogeneous PIM-based NoC architectures, integrating more comprehensive pipeline modeling. Additionally, HPIM-NoC features an architecture search framework targeting heterogeneous PIM core configurations, utilizing priori knowledge to further accelerate the search process. The contributions include:

- The simulator implemented in HPIM-NoC overcomes the constraints of homogeneous simulators that cannot model heterogeneous core configurations. A heterogeneous architectural prototype is defined and a workload allocation algorithm is introduced for heterogeneous cores. HPIM-NoC models the data transmission in neural networks and enables the simulation of pipelining between core computation and NoC transmission.
- A novel search framework is proposed by employing a customized simulated annealing (SA) [17] algorithm with specific operators and reward function targeting for heterogeneous architectures. A three-stage search process is designed based on priori knowledge, effectively addressing the challenge of the vast search space in heterogeneous architectures. Additionally, a fine-tuning method is introduced to ensure the validity of architectural configurations throughout the three-stage search process. The search process is accelerated by precomputing core PPA metrics and reducing NoC simulation frequency to minimize redundant computations. Furthermore, a customized placement algorithm is designed to optimize heterogeneous NoC layouts, reducing communication latency and overall area.
- Experimental results demonstrate that the Fusions of Metrics (FoMs) of heterogeneous PIM-based NoC architectures found by HPIM-NoC are reduced by 1.18%, 16.94%, and 37.41% for ResNet-18 than homogeneous architectures under the three settings, respectively. The layouts of heterogeneous PIM-based NoC are also obtained with the shortest total wirelength and optimized total area. The results demonstrate the effectiveness of HPIM-NoC in quickly exploring the near-optimal heterogeneous PIM-based NoC architecture within a limited time.

II. RELATED WORKS

Currently, PIM simulators play an important role in the design and optimization of neural network accelerators. DNN+NeuroSIM v2.0 [10] and MNSIM 2.0 [11] provide circuit-level and architecture-level modeling homogeneous PIM-based multi-array architectures with various PIM types. However, these works lack the simulation of data transmission on the NoC and pipeline between computation and transmission, leading to potential deviations in simulation results. SIAM [18] provides simulation for homogeneous PIM-based NoC architectures with different mapping schemes and parameters. All of the above simulators are designed for homogeneous PIM-based architectures and cannot support the simulation of heterogeneous PIM-based NoC architectures.

In the field of NoC simulation, Booksim 2.0 [12] is widely used to evaluate the performance of general-purpose NoC architectures. Booksim uses the injection rate to represent the

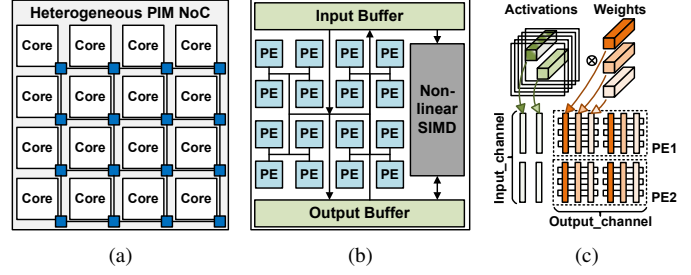


Fig. 1. The prototype of heterogeneous PIM-based NoC architecture for simulation. (a) The heterogeneous PIM-based NoC accelerator. (b) The core in heterogeneous PIM-based NoC. (c) The partitions of activations and weights.

number of packets transmitted per unit of time. During the simulation, random numbers are continuously generated. If the random number exceeds the injection rate, a packet is sent along the designated path, and the transmission latency from sending to receiving is recorded. However, the computational characteristics of neural networks introduce inter-layer data dependencies. The differences in dataflows between general-purpose NoCs and specific NoCs for neural networks lead to deviations in the simulation results of Booksim.

An architecture search framework provides an efficient and systematic method for exploring the vast design space, which is crucial for guiding the design of neural network accelerators. PIM-HLS [13] proposes a dynamic weight scheduling algorithm for scheduling and mapping across heterogeneous PIM arrays. Additionally, [14] optimizes mapping and scheduling strategies by defining communication priorities and NoC communication behavior tables. However, these works only explore mapping and scheduling strategies, without searching for hardware configurations like the PIM types and array sizes. Gibbon [15] automatically searches for the optimal hardware parameters and network structures for homogeneous PIM-based designs. AIG-CIM [16] employs a design space exploration framework with only three different configurations of SRAM-based PIM arrays for heterogeneous PIM NoC accelerator. Therefore, a specific search framework is still lacking to guide the design of heterogeneous PIM-based NoC architectures.

III. THE DESIGN OF HPIM-NoC

HPIM-NoC consists of a simulator (Section III-A) and a search framework (Section III-B) for heterogeneous PIM-based NoC architectures. Heterogeneous simulation is achieved by calculating the PPA of each core separately and comprehensive pipeline modeling. Based on the heterogeneous NoC simulator, the architecture search framework of HPIM-NoC adopts a priori-based three-stage search approach to reduce the complexity of the heterogeneous search process.

A. Heterogeneous PIM-based NoC Simulation

Existing simulators lack the integration of simulation of heterogeneous cores and NoC transmission, thus failing to provide accurate performance evaluation for heterogeneous PIM-based NoC architectures. A novel simulator is essential to address the challenges of workload allocation, inter-core data transmission, and precise simulation of neural network

TABLE I
CONFIGURABLE PARAMETERS OF THE PROTOTYPE

Configuration Level	Configurable Parameters	Options
Algorithm	Neural Network	Network structure, layer size, etc.
PIM Core	PIM_Type	SRAM, RRAM
	PE_Num	$1 \times 1, 2 \times 2$, etc.
	Array_Size	$32 \times 32, 64 \times 64$, etc.
NoC	Topology	Mesh, Cmesh, etc.
	Core_Num	$1 \times 1, 2 \times 2$, etc.
Mapping	Mapping order	Normal, snake, zigzag

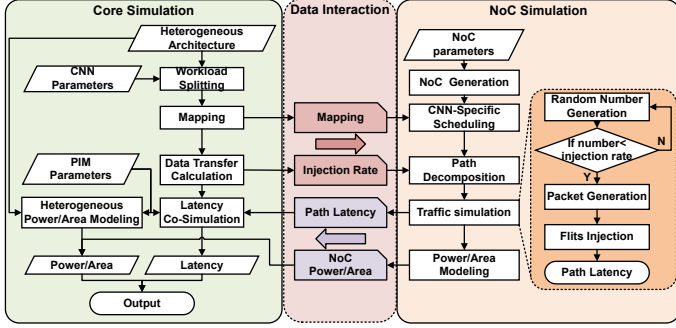


Fig. 2. The proposed simulation flow chart of HPIM-NoC.

data interactions. Before the implementation of the simulator, the prototype of heterogeneous PIM-based NoC architecture is essential to be defined first. As shown in Fig. 1, the prototype architecture can be divided into three levels, including the overall neural network accelerator (Fig. 1 (a)), the heterogeneous PIM core (Fig. 1 (b)), and the PE containing the PIM arrays and peripheral circuits. All configurable parameters of the architecture prototype are listed in Table I. As shown in Fig. 2, the proposed simulator of HPIM-NoC consists of three main components: intra-core simulation, NoC data transmission simulation, and data interaction. Based on the configurations of neural networks and heterogeneous architectures, the workload mapping and inter-core data transmission paths are generated. The PPA results of interconnection circuits are obtained by the NoC simulation, and then jointly simulated with the simulation results of each heterogeneous PIM core.

To simulate the heterogeneous PIM cores accurately, the core simulation must be flexible enough to account for various PIM types and configurations. The core simulation supporting heterogeneous PIM architectures is implemented based on the homogeneous PIM simulator MNSIM [11]. The partition of Weights and activations according to the cores and PEs is shown in Fig. 1 (c), then the data transfer volume between cores is obtained. Differences between the capacity of cores prevent the direct application of load-balancing methods from homogeneous systems to heterogeneous architectures. To address this, a backtracking algorithm (Algorithm 1) is designed to solve the workload allocation and mapping challenges for heterogeneous cores. Simulations are performed on each core and PPA results are obtained.

Existing NoC simulators, such as Booksim [12], do not support the simulation of CNN-specific inter-layer data transmission. An NoC simulator has been proposed based on Booksim to model the inter-layer data transmissions in CNN. The

Algorithm 1 Backtracking to Allocate Cores for Workload

Input: List of computing cores, workload size $W \times H$.

Output: Indices of selected cores.

```

1: Initialize a list of core usage, and an empty result list;
2: function Backtrack()
3:   Find the first empty spot in the workload grid;
4:   if no empty spot then
5:     return True    ▷ All workload is stored in cores
6:   end if
7:   for core  $c_i$  in unused cores do
8:     if core  $c_i$  can store workload at the empty spot then
9:       Place  $c_i$  in the grid and mark it as used;
10:      Append  $i$  to the result list;
11:      if Backtrack() then    ▷ Backtracking
12:        return True
13:      end if
14:      Remove  $c_i$  and unmark it as used;
15:    end if
16:  end for
17:  return False
18: end function
19: if Backtrack() then
20:   return result list
21: end if

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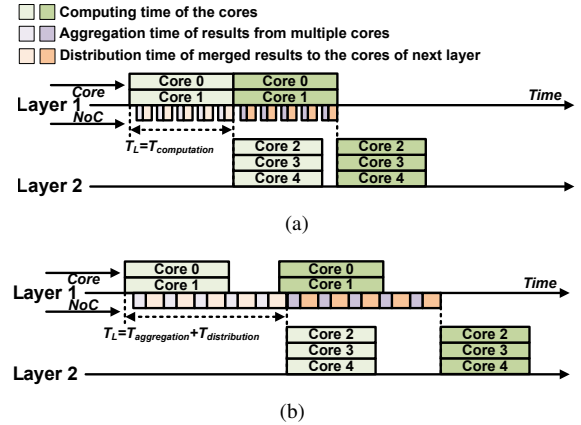


Fig. 3. The pipeline of computation in the cores and data transmission in the NoC. (a) The calculation time is larger than the transmission time. (b) The calculation time is smaller than the transmission time. Different shades of colors correspond to multiple sets of data.

injection rate for path i is calculated by the volume of data transmitted per second and the total bandwidth, as Equation 1:

$$\lambda_i = \frac{A_i \times N_{bits} \times FPS}{C_i \times W}, \quad (1)$$

where A_i represents the number of activations, N represents the data precision, FPS represents the number of frames per second, C_i represents the number of cores to receive the data, and W represents the bandwidth of one path.

The integration of core computation and NoC data transmission is proposed to enable a holistic simulation for both the processing and communication stages. The aggregation latency in each layer and the distribution latency between two adjacent layers in the neural network are both obtained by the maximum latency among multiple paths. The computation of data in cores

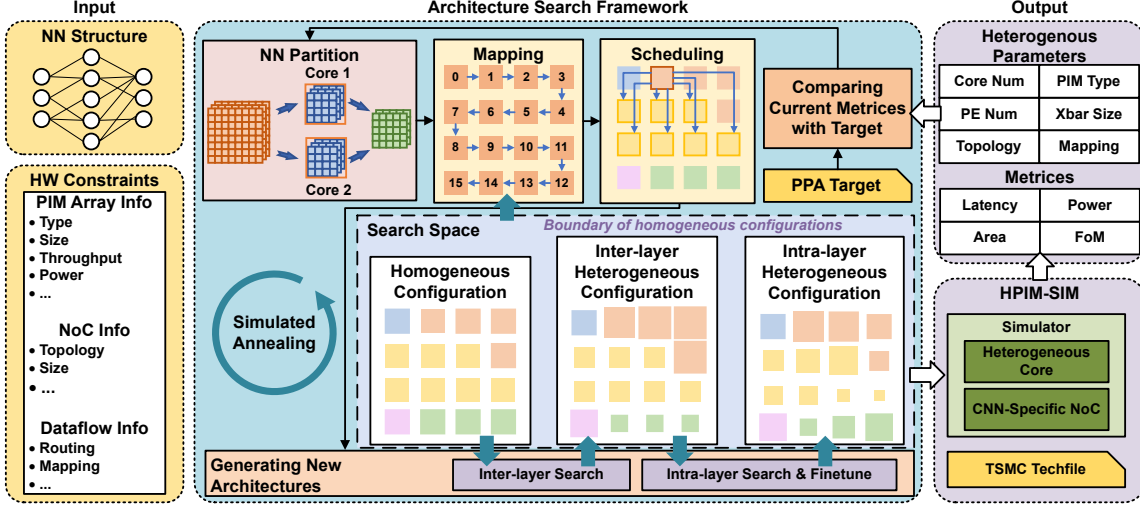


Fig. 4. The proposed search framework of HPIM-NoC.

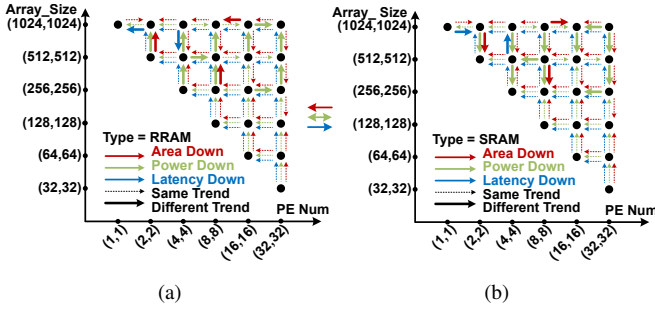


Fig. 5. The changing trend of the PPA results of each core with different configurations. (a) RRAM-based PIM core. (b) SRAM-based PIM core. Only a few RRAM-based and SRAM-based configurations have different trends.

and the transmission in NoC can be performed as the pipeline shown in Fig. 3, while previous simulators [10], [11] default to sequential computation and transmission. The total latency of each neural network layer is equal to the larger value of the calculation time for cores and the transmission time for NoC.

B. Heterogeneous PIM-based NoC Architecture Search

Despite the heterogeneous PIM-based NoC simulator's ability to quickly obtain PPA results for specific architectural configurations, the numerous configurable parameters make it difficult to manually design an NoC that meets the target PPA efficiently. Existing search frameworks designed for homogeneous NoC architectures are unable to effectively navigate the vast design space of heterogeneous architectures. A novel search framework for heterogeneous PIM-based NoC architectures has been proposed based on a customized simulated annealing (SA) algorithm, as shown in Fig. 4.

Due to the need for individual configuration of each computing core in a heterogeneous PIM-based NoC, the search space is significantly larger than that of homogeneous architectures. As the number of cores increases, the exponential growth in configuration possibilities makes it computationally expensive to explore all potential configurations. For example, the number of possible heterogeneous configurations with 16×16 cores can reach up to 10^{476} ($N_{Mapping} \times (N_{PIM_Type} \times N_{PE_Num} \times$

$N_{Array_Size})^{16 \times 16}$). The enormous search space requires efficient methods to narrow down configurations to those that are most likely to meet target PPA requirements.

Given that heterogeneous architectures are composed of various cores from homogeneous architectures, the proposed HPIM-NoC search framework adopts a priori-knowledge-based approach, utilizing a three-stage search process to narrow the search space and quickly identify a viable architectural solution. The framework first identifies the homogeneous configuration closest to the PPA target as the starting point for further exploration. Since the PPA trends of cores are not strictly linear with respect to Array_Size and PE_Num, the PPA results and variation trends for cores are precomputed, as illustrated in Fig. 5. Subsequently, the search framework proceeds to explore inter-layer heterogeneous architectures by changing cores according to the trends in Fig. 5, applying the same configuration to each PIM core for a given neural network layer. For example, if the current homogeneous configuration (RRAM-based PIM, Array_Size=(256, 256)) gets higher latency than the target PPA, the cores for a specific layer can be adjusted to SRAM-based PIM or Array_Size can be increased to (512, 512) following the trend in Fig. 5. If the desired PPA target remains unmet, the search process continues to investigate intra-layer heterogeneous architectures, employing different configurations for the PIM cores within each layer. The three-step strategy drastically reduces the size of the search space at each stage, speeding up the process while ensuring the effectiveness of the search results. For example, the search space for heterogeneous configurations with 16×16 cores is reduced from 10^{476} to a range of 10^{21} to 10^{43} (largest single-stage search space among the three stages), significantly accelerating the search process.

Then, a specific SA algorithm is designed for heterogeneous PIM-based NoC architectures with a set of operators. Four operators are proposed to modify configurations within the same layer, including changes to the PIM type, PE number, and array size, as well as adjusting the core order using a new mapping scheme. Additionally, the intra-layer search introduces an extra operator to swap two cores within the

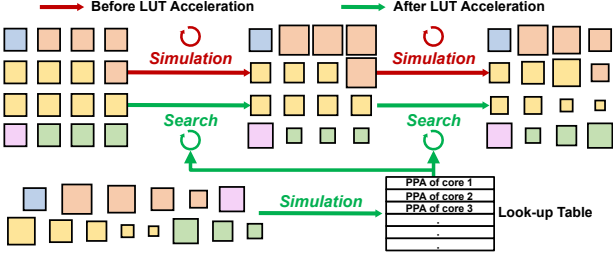


Fig. 6. The searching process before and after acceleration.

same layer. During the intra-layer search process, changing the size of the cores within a given layer may result in the number of weights exceeding the total available capacity. To address this, a method is proposed to finetune the heterogeneous PIM cores within a layer after each operation to ensure the feasibility of the configuration. The total capacity of the cores corresponding to the current neural network layer is calculated, and the number of weights is checked. If the weights exceed the available capacity, additional cores are allocated during inter-layer heterogeneity for that layer. If the weights fit within the capacity but the difference exceeds the smallest core's capacity, the smallest core is removed.

Since the PPA results include three metrics, a composite metric Fusion of Metrics (FoM) is proposed as Equation 2:

$$\text{FoM} = A^\alpha \times P^\beta \times L^\gamma, \quad (2)$$

where A , P , and L represent area, power, and latency respectively. α , β , and γ are user-defined parameters that control the importance assigned to different metrics during the search process. The simulated annealing algorithm employs FoM as the feedback indicator in the search process. The number of operations performed in the SA algorithm decreases with the number of steps. The search concludes when the target PPA is achieved or the search steps are completed.

Due to the simulation required by each step in the search process, the simulation time becomes unacceptable after multiple iterations. Each simulation will re-simulate all cores of the current heterogeneous architecture to obtain PPA metrics, significantly slowing down the overall search speed. To address this, an acceleration method based on the look-up table (LUT) for simulation is proposed. As shown in Fig. 6, the red arrows represent the original search and simulation process, while the green arrows indicate the accelerated process using the lookup table. All possible core configurations are simulated beforehand, and the PPA results are stored. During the search, the pre-computed results are reloaded and accumulated, eliminating redundant core simulations. After accelerating core simulations, NoC simulations become the most time-consuming part of the search process. To further optimize, the frequency of NoC simulations is reduced during the search process by leveraging the trend of data transmission patterns between cores gradually stabilizing during the search process. Finally, a heterogeneous PIM-based NoC placement algorithm is implemented using a force-directed approach. The algorithm ensures a compact arrangement of heterogeneous cores by leveraging the attractive forces along the topological interconnections.

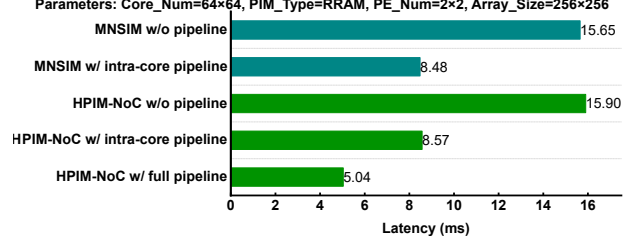


Fig. 7. The simulation results of MNSIM and the proposed HPIM-NoC with different pipeline modes for VGG-8.

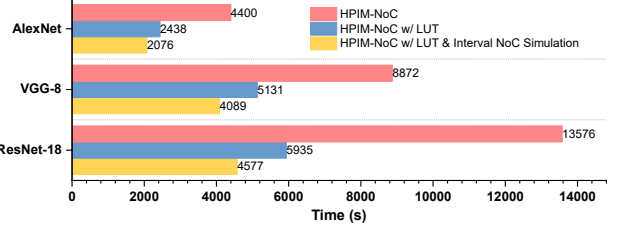


Fig. 8. The search times of the proposed HPIM-NoC with different acceleration methods for ResNet-18 under $(\alpha = 1, \beta = 1, \gamma = 1)$. The search step is set to 100 and Core_Num is 24×24 for a fair comparison.

IV. EXPERIMENTAL RESULTS

First, the correctness of the proposed simulator and co-simulation process is validated by comparing it with MNSIM through simulations of homogeneous architectures. MNSIM supports two modes: non-pipeline and intra-core pipeline. The proposed HPIM-NoC simulator supports three modes: non-pipeline, intra-core pipeline, and full pipeline between computation and transmission. The latency results under different pipelining conditions were obtained, as shown in Fig. 7. The results of the HPIM-NoC simulator show only slight differences from MNSIM in the first two modes, indicating the correctness of the HPIM-NoC simulation. The difference arises because MNSIM does not account for the NoC data transmission latency. In the full pipeline mode, the simulator HPIM-NoC gets lower latency than in the intra-core pipeline mode, demonstrating the effectiveness of the co-simulation.

In the search framework, the parameters for the simulated annealing algorithm are set as follows: initial temperature $T_0 = 100$, final temperature $T_f = 0$, and cooling rate $\alpha = 0.99$. The number of operations per step is set as $\text{random}(0, 1) \times \frac{T}{T_0} + 1$. We tested the search performance under three settings: $(\alpha = 0, \beta = 0, \gamma = 1)$ for improved latency, $(\alpha = 0, \beta = 1, \gamma = 1)$ for better energy-delay product, and $(\alpha = 1, \beta = 1, \gamma = 1)$ for an optimized customized FoM. AlexNet, VGG-8, and ResNet-18 are selected as the CNN models to evaluate HPIM-NoC, while other models are also applicable. ImageNet is selected as the dataset. The acceleration methods tailored for the search process are first tested, with the time of searching architectures for three networks shown in Fig. 8. The proposed acceleration method reduces the search time of HPIM-NoC by $2.12\times$, $2.17\times$, and $2.96\times$, respectively.

The search results for the three networks are shown in Table II. Due to the inherent randomness of the search process, the resulting topologies, core counts, and configurations of each core vary under different search parameters. HPIM-NoC effec-

TABLE II
THE SEARCH RESULTS OF THE PROPOSED HPIM-NOC FOR DIFFERENT NEURAL NETWORKS.

Neural Network	Search Method	Topology	Core_Num	Area (mm ²)	Power (W)	Latency (ms)	Time (s)
AlexNet	($\alpha = 0, \beta = 0, \gamma = 1$)	Mesh	13×13	98.95	21.45	0.24	2180
	($\alpha = 0, \beta = 1, \gamma = 1$)	Mesh	11×11	62.89	11.97	0.27	4322
	($\alpha = 1, \beta = 1, \gamma = 1$)	Mesh	10×10	73.49	12.99	0.28	3749
VGG-8	($\alpha = 0, \beta = 0, \gamma = 1$)	Cmesh	16×16	179.09	44.12	1.59	4365
	($\alpha = 0, \beta = 1, \gamma = 1$)	Mesh	16×16	112.08	21.05	1.51	6990
	($\alpha = 1, \beta = 1, \gamma = 1$)	Mesh	14×14	129.45	26.32	1.51	6559
ResNet-18	($\alpha = 0, \beta = 0, \gamma = 1$)	Cmesh	22×22	357.35	86.57	0.98	4936
	($\alpha = 0, \beta = 1, \gamma = 1$)	Mesh	26×26	321.23	57.72	0.97	10882
	($\alpha = 1, \beta = 1, \gamma = 1$)	Mesh	24×24	288.69	55.44	0.90	11110

TABLE III
THE SEARCH RESULTS OF THE PROPOSED HPIM-NOC FOR RESNET-18 WITH ($\alpha = 1, \beta = 1, \gamma = 1$)

PIM_Type	Search Method	Area (mm ²)	Power (W)	Latency (ms)	FoM (10 ¹⁶)	Improvement	Time (s)
SRAM-based	Homogeneous config w/o search	383.01	82.38	1.05	3.32046	/	/
RRAM-based	Homogeneous config w/o search	340.32	63.43	1.06	2.29240	Baseline	/
Heterogeneous	Inter-layer heterogeneous	317.98	60.65	0.92	1.77191	-22.71%	6658
	Intra-layer heterogeneous	288.69	55.44	0.90	1.43500	-37.41%	11110
	Optimal heterogeneous config	288.69	55.44	0.90	1.43500	-37.41%	∞

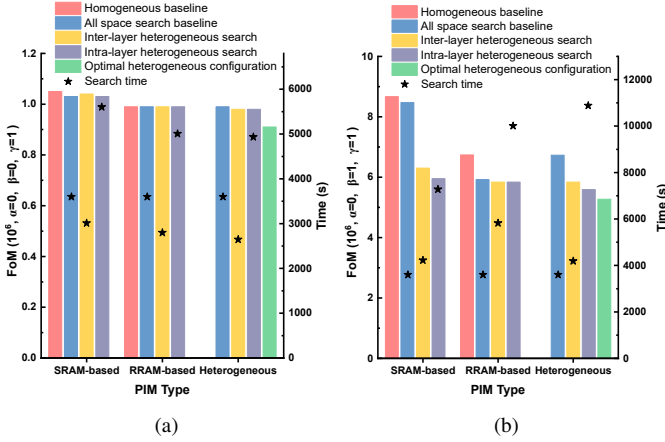


Fig. 9. The search results of the proposed HPIM-NoC for ResNet-18 with (a) ($\alpha = 0, \beta = 0, \gamma = 1$) and (b) ($\alpha = 0, \beta = 1, \gamma = 1$).

tively identifies heterogeneous NoC architectures optimized for the target PPA, showcasing distinct tendencies across various performance metrics. The detailed results for the first two settings for ResNet-18 are shown in Fig. 9 (a) and (b), while the detailed results for ResNet-18 with the third setting are presented in Table III. Homogeneous PIM-based NoC architectures and full-space search results were used as baselines for comparison. The results show that full-space search fails to identify solutions with effective improvement within the limited given time. Under the three settings for ResNet-18, the FoMs of heterogeneous PIM-based NoC architectures found by HPIM-NoC are reduced by 1.18%, 16.94%, and 37.41% compared to homogeneous architectures, respectively. The heterogeneous NoC layouts for ResNet-18 with three configurations are shown in Fig. 10, achieving the shortest total wirelength and optimized total area. By minimizing the distance between cores based on the topological interconnections during the placement process, the layouts achieve virtually no crossover points in subsequent routing. The results demonstrate the effectiveness of HPIM-NoC in exploring the near-optimal heterogeneous PIM-based NoC architectures, achieving significant performance improvements compared to homogeneous architectures.

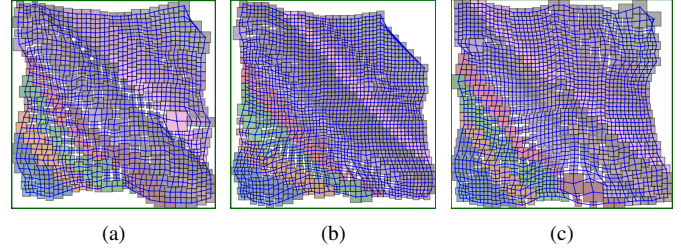


Fig. 10. The layout examples of (a) ($\alpha = 0, \beta = 0, \gamma = 1$) with Core_Num = 22 × 22, (b) ($\alpha = 0, \beta = 1, \gamma = 1$) with Core_Num = 26 × 26, and (c) ($\alpha = 1, \beta = 1, \gamma = 1$) with Core_Num = 24 × 24 for ResNet-18. Different colors of cores represent mapping to different neural network layers. The lines represent topological interconnections between cores.

V. CONCLUSION

In this work, we propose HPIM-NoC, a comprehensive simulation and architecture search framework specifically designed for heterogeneous PIM-based NoC architectures. HPIM-NoC employs a modified simulated annealing algorithm with a three-stage search process, and a fine-tuning method is introduced to ensure the validity of the discovered architectures. Experimental results demonstrate that the accelerated HPIM-NoC achieves over a 10× reduction in search time during the same exploration process. The FoMs of heterogeneous PIM-based NoCs found by HPIM-NoC are reduced by 1.18%, 16.94%, and 37.41% for ResNet-18 under three settings, respectively, compared to homogeneous architectures. HPIM-NoC is capable of quickly finding near-optimal configurations, providing a powerful tool for heterogeneous PIM-based NoC designs.

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